

Research Article

Development of a Monoclonal Antibody Sandwich ELISA for the Quality Control of Human and Animal Tetanus Vaccines

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Abstract

Antigen identity, quantity and integrity are key factors to be evaluated as part of consistency testing of tetanus vaccines. Here we have developed a monoclonal antibody sandwich ELISA to measure the relative amount and quality of tetanus toxoid (TTxd) in human and animal tetanus vaccines. The ELISA is highly specific, has good dilutional linearity, and is suitable for detecting TTxd in a range of different products. We have demonstrated the ability of the assay to discriminate between batches of different content, using vaccine batches that had been prepared to contain differing amounts of TTxd, and of different quality, using samples of non-adjuvanted TTxd that had been exposed to sonication and final lot vaccines that had been exposed to heat or oxidative stress. We have also demonstrated successful transfer of the method to other laboratories and have shown that different tetanus antigen materials may be able to serve as a reference antigen for standardization of the method. The results show this test has the potential to play a key role in a control strategy no longer including an *in vivo* potency test.

Plain language summary

Tetanus vaccines help to protect against tetanus infection. Currently, animal tests are used to ensure the potency of such vaccines. Since these tests were first introduced, there have been improvements in non-animal technologies that can be used to ensure consistent production of potent vaccine batches. To demonstrate that a new batch of tetanus vaccine is consistent with a previous batch of known potency, the quality and amount of the component that stimulates the immune response upon vaccination must be assessed in comparison. We have developed an assay that can measure the quality of a range of different tetanus vaccine product types. The assay is very specific and reliable, and different laboratories obtained comparable results, showing that the assay is suited for routine use. Once validated by manufacturers and accepted by regulators, this assay will greatly reduce the number of animals needed for batch release of tetanus vaccines.

1 Introduction

Every vaccine batch that is manufactured must undergo a series of rigorous tests to ensure it meets certain safety and potency standards. Traditionally, animal testing has played a major role in routine quality control testing of vaccines, and many animals are still used in Europe for this purpose (EC, 2020). Over the last few decades there has been a strong drive to review the state of the science of human and veterinary vaccine potency and safety testing methods, and to identify opportunities to advance new and improved methods that can replace, reduce, and refine animal use (3Rs) (McFarland et al., 2011; Kulpa-Eddy et al., 2011; Schutte et al., 2017; Akkermans et al., 2020).

The consistency approach is one concept that offers an alternative to the current quality control strategy for established vaccines (De Mattia et al., 2011). Since vaccines were first developed, there have been significant improvements in the vaccine production process, resulting in less batch-to-batch variation and a lower risk of producing unsafe or ineffective products. Rather than each new batch of vaccine being seen as unique, it can be considered as one of a series of batches that shares many of the same characteristics as a manufacturer-specific vaccine batch of proven clinical efficacy and safety. Monitoring the key qualitative and quantitative attributes of a vaccine with a panel of *in vitro* methods should demonstrate that final batches are of consistent quality, making the use of an inherently variable *in*

vivo potency test on the final product unnecessary (Council of Europe, 2018).

The VAC2VAC consortium¹, funded by the Innovative Medicines Initiative (IMI2) program, was initiated in 2016 with the aim of developing new tests that support the use of the consistency approach in batch testing of established vaccines. Tetanus was selected as one of the model vaccines for this project based on the large proportion of animals used for potency testing and the severity of discomfort of the potency assays. Vaccines for human and animal use face the same challenges when changing from *in vivo* to *in vitro* methods, and the VAC2VAC project brought together both human and veterinary tetanus vaccine manufacturers, providing a unique opportunity for cross-collaboration of these two areas of medicine.

The tetanus toxoid (TTxd) vaccine was developed by Ramon and collaborators in the early 1920s and used extensively for the first time among soldiers during World War II (Scheibel, 1955; Levy, 1975). Since then, immunization programs using TTxd-containing vaccines have been highly successful in preventing maternal and neonatal tetanus as well as injury-associated tetanus (WHO, 2017). The incidence of tetanus in animals has also decreased due to consistent immunization programs. Tetanus vaccines for human use consist of formaldehyde-detoxified tetanus toxin (TTxd) adsorbed onto an aluminium hydroxide (Al(OH)₃) and/or an aluminium phosphate (AlPO₄) adjuvant in combination with additional non-tetanus components to protect against other vaccine preventable diseases. For immunization of children, tetanus is given as a combined diphtheria-tetanus-pertussis (DTP) vaccine or diphtheria-tetanus-acellular pertussis (DTaP) vaccine. These combination vaccines may also include inactivated poliomyelitis virus (IPV), *Haemophilus influenzae* type B capsular polysaccharide (Hib), and hepatitis B surface antigen (HepB) components. For booster dosing of adults and adolescents, a tetanus-diphtheria combination with a lower concentration of diphtheria antigen (d) is available. Vaccines for livestock are available as monovalent tetanus vaccines, bivalent influenza and tetanus vaccines, or multivalent vaccines combining toxoids produced by different *Clostridium* species and antigens protecting against pulmonary disease. In addition to the Al(OH)₃ and AlPO₄ adjuvants found in human vaccines, veterinary tetanus vaccines may contain potassium aluminium sulfate (KAl(SO₄)₂) as well as non-aluminium adjuvants including carbomer- and saponin-based adjuvants (e.g., Carbopol and Quil A).

Potency of tetanus vaccine is traditionally determined in international units (IU) of protection using an *in vivo* challenge test where survival of immunized guinea pigs or mice is assessed following challenge with tetanus toxin (WHO, 1990; Council of Europe, 2008). For the non-vaccinated or only partly protected animals, the challenge with tetanus toxin results in severe suffering. In the context of the 3Rs, efforts have been made to replace challenge assays by serology assays where the blood of vaccinated test animals is collected, and antigen-specific antibody titers are determined (Hendriksen et al., 1994; Hendriksen and Winsnes, 2002; Stickings et al., 2011). Potency assays, both challenge and serology assays, raise ethical concerns because of animal suffering and/or the very large

number of animals used in the process, and are also expensive and time consuming. Furthermore, such assays have poor precision and poor reproducibility (Stalpers et al., 2021), raising questions about their relevance. Although further efforts have been made to replace these *in vivo* tests entirely with an *in vitro* alternative, to date, no non-animal method for tetanus vaccine potency testing has gained regulatory acceptance.

The quantity and quality of antigen present in tetanus vaccines is of utmost importance for product safety and effectiveness and is therefore a critical factor to be evaluated as part of consistency testing. We have previously developed a sandwich enzyme-linked immunosorbent assay (ELISA) for the quantification of TTxd in combination vaccines and showed that this method was suitable for monitoring the consistency of tetanus antigen in a range of different vaccines licensed for human use (Coombes et al., 2012). As part of the VAC2VAC project we further demonstrated that this ELISA was suitable for characterizing TTxd in a range of veterinary tetanus vaccines and could potentially be used as part of a control strategy for these products (Riches-Duit et al., 2019). However, the original ELISA that was developed used a monoclonal antibody (mAb) for capture of the TTxd and a polyclonal antibody (pAb) for detection, and it is recognized that pAbs are harder to implement as part of a control strategy due to their batch-to-batch variability and differences in antibody reactivity. In addition, pAbs require the use of animals every time a new batch is produced. We therefore developed a new assay that utilizes well-characterized mAbs for both capture and detection of the TTxd. These antibodies were selected based on their capability to bind the native, detoxified, and adsorbed antigen, and altered antigen following exposure to elevated temperature. Epitope competition studies were also performed to identify a pair of high affinity antibodies that could be used in a sandwich ELISA format (Riches-Duit et al., 2021). Here we have evaluated assay performance characteristics of the mAb ELISA (within and between laboratories), demonstrated suitability for use with different product types, and demonstrated the ability of the assay to discriminate between vaccine batches of different quality.

2 Materials and methods

2.1 Monoclonal antibodies

The two anti-tetanus mAbs selected for ELISA development were TT010 (a rat IgG antibody provided by the National Institute for Biological Standards and Control, NIBSC, UK) and 8E1-1H1.2.1 (a mouse IgG antibody provided by Sanofi). The mAbs were produced by hybridoma culture and purified by Protein A/G affinity chromatography and buffer exchanged into PBS for storage. Stocks of these purified mAbs are available from the NIBSC catalogue².

2.2 Toxoid and vaccine samples

Toxoid and vaccine samples were obtained from two human vaccine manufacturers (coded HuA and HuB) and four veterinary vaccine manufacturers (coded A, B, C and D) in the VAC2VAC

¹ <https://europevaccine.wixsite.com/vac2vac-eu>

² <https://nibsc.org/>

**Tab. 1: Summary of drug substances and drug products tested**

Manufacturer	Site	Product/drug substance	Labelled concentration of TTxd (Lf/mL)	Adjuvant
HuA	1	TTxd	2300	N/A
		dT-IPV	20	Al(OH) ₃
		DTaP-IPV	20	Al(OH) ₃
		DTaP-IPV-HepB-Hib ^a	20	Al(OH) ₃
	2	TTxd	2280	N/A
		TTxd adsorbed	270	AlPO ₄
		dTaP ^{b,c}	10	AlPO ₄
		DTaP-IPV	10	AlPO ₄
HuB	N/A	TTxd	4100	N/A
		DT adsorbed	67	Al(OH) ₃
		DTaP ^{a,b,c,d}	20	Al(OH) ₃
		DTaP-IPV	20	Al(OH) ₃
		DTaP-HepB-IPV	20	Al(OH) ₃ + AlPO ₄
		dTaP ^a	10	Al(OH) ₃ + AlPO ₄
		dTaP-IPV	10	Al(OH) ₃ + AlPO ₄
Veterinary company A	N/A	A / TTxd	Not reported	N/A
		A / Monovalent ^a	Not reported	KAl(SO ₄) ₂
		A / Trivalent	Not reported	KAl(SO ₄) ₂
Veterinary company B	N/A	B / TTxd 1	Not reported	N/A
		B / TTxd 2	Not reported	N/A
		B / TTxd 3	Not reported	N/A
		B / TTxd 4	Not reported	N/A
		B / Trivalent	Not reported	Al(OH) ₃
		B / Multivalent 1 ^{a,d}	Not reported	Al(OH) ₃
		B / Multivalent 2 ^b	Not reported	KAl(SO ₄) ₂
		B / Monovalent 1 ^{b,c}	Not reported	Saponin based
		B / Monovalent 2 ^b	Not reported	Carbopol based
Veterinary company C	N/A	C / TTxd	Not reported	N/A
		C / Multivalent ^b	Not reported	Carbomer
Veterinary company D	N/A	D / TTxd 1	Not reported	N/A
		D / TTxd 2	Not reported	N/A
		D / TTxd 3	Not reported	N/A
		D / Monovalent	Not reported	AlPO ₄
		D / Bivalent	Not reported	AlPO ₄ , Quil A, ISCOMS

^a Products used in the graded dose studies, ^b Products used to assess the ability of the ELISA to detect temperature-induced changes, ^c Products used to assess the ability of the ELISA to detect chemical-induced changes, ^d Products used in the qualification studies

consortium. The samples from the human manufacturers included upstream TTxd and adsorbed TTxd drug substances together with a range of pediatric (DTP and DTaP based) and booster (dT based) dose drug products. Similarly, the samples from the veterinary manufacturers included a selection of non-adjuvanted TTxds used for vaccine formulation and a wide range of final lot vaccines. See Table 1 for a summary of drug substances and products tested.

Where available, the labelled tetanus antigen concentration in flocculating (Lf) units is shown.

Drop out samples without any tetanus antigen were received from HuA (dTaP and DTaP-IPV-HepB-Hib), HuB (dTaP and DTaP), veterinary company A (A/ Monovalent) and veterinary company B (B/ Multivalent 1). For the human vaccines, drop out samples covered all adjuvant types (Al(OH)₃ and AlPO₄ for HuA

and $\text{Al}(\text{OH})_3$ and $\text{Al}(\text{OH})_3 + \text{AlPO}_4$ for HuB) and all potential additional non-tetanus antigens found in combination vaccines. For the veterinary vaccines, drop out samples covered the two aluminium adjuvant types used ($\text{Al}(\text{OH})_3$ and $\text{KAl}(\text{SO}_4)_2$), other clostridial toxoids (*Cl. perfringens* beta toxoid, *Cl. perfringens* epsilon toxoid, *Cl. septicum* toxoid, *Cl. chauvoei* toxoid) as well as *Mannheimia haemolytica* and *Pasteurella trehalose* bacterial cell preparations. Drop out samples were used to confirm specificity of the ELISA assay for TTxd.

The 3rd WHO International Standard (IS) for Tetanus Toxoid for use in Flocculation Test (NIBSC code 16/302, 970 Lf/ampoule) and 4th WHO IS for Tetanus Toxoid (Adsorbed) (NIBSC code 08/218, ~25 Lf/ampoule) were included in some assays, either as a positive control or as part of studies to determine the suitability of different tetanus toxoid-containing material to act as a reference preparation in the ELISAs. Each ampoule of 16/302 or 08/218 was reconstituted in 1 mL of ultrapure water to give a concentration of 970 Lf/mL and approximately 25 Lf/mL, respectively.

2.3 Desorption studies

Aluminium adjuvanted final lots were centrifuged (13,000 rpm for 10 min) to separate them into non-adsorbed (vaccine supernatant) and adsorbed (adjuvant pellet) fractions. The pellets were re-suspended in a sodium phosphate/EDTA solution (1 volume of 56 g/L EDTA: 49 volumes 90 g/L disodium hydrogen phosphate) and incubated for 16–20 h at +37°C to elute the adsorbed antigen. Finally, a second centrifugation step was performed to remove any remaining adjuvant. The desorbed samples were titrated in the ELISA alongside the non-adsorbed fraction (supernatant) and the whole (adjuvanted) vaccine sample. Relative antigen estimates for the adsorbed fraction and non-adsorbed fraction were calculated relative to the whole vaccine sample.

2.4 Graded dose samples and spiking assays

Graded dose samples were prepared for the final vaccine products for which drop out samples had been received by mixing them with the normal vaccine sample (100% TTxd) in different ratios to give samples containing 25%, 50% and 75% tetanus antigen. To allow over-formulated samples to be prepared, one of the drop out samples from HuA was also spiked with adsorbed TTxd (270 Lf/mL) from the same manufacturer to give 25%, 50%, 75%, 100%, 200% and 400% tetanus antigen.

2.5 Alteration of non-adjuvanted TTxd by sonication

TTxd from HuB was diluted to 20 Lf/mL and subjected to different sonication conditions. Sonication was performed using a 4 mm diameter sonication tip of a Sanyo Soniprep submerged in the TTxd sample, which was continuously cooled on ice. An increasing number of sonication cycles of 15 sec with a 15 sec interval and an amplitude of 15 micron were performed (T0–T21). A lower energy input sample was generated with 21 cycles using an amplitude of 7.5 instead of 15 micron (T21 50%), and a longer interval cooling sample was generated by performing 21 cycles with a 45 sec interval (T21 extra cool).

Altered toxoid samples were analyzed by size exclusion chromatography on a Superose12 column (Invitrogen) with PBS as the

eluent at 0.5 mL/min and UV detection at 280 nm, using an Agilent 1100 series HPLC system.

2.6 Alteration of vaccine samples by heat stress

Several vaccine products from four different manufacturers were subjected to heat stress studies. Samples were incubated for 8 weeks at elevated temperatures of +37°C and +45°C, with control samples held at the normal storage temperature of +4°C. The aluminium adjuvanted vaccine samples were tested in the ELISA with and without desorption (to determine if the adsorption profile was also changing after incubation at elevated temperatures), as described in Section 2.3. The whole vaccine control sample at +4°C was included on every plate to act as a reference. The reference was tested in duplicate columns, and all other samples were tested in a single column.

2.7 Alteration of vaccine samples by oxidative stress

Vaccine products from HuA (dTaP, AlPO_4 adjuvant), HuB (DTaP, $\text{Al}(\text{OH})_3$ adjuvant), and veterinary company B (B/ Monovalent 1, saponin-based adjuvant) were subjected to oxidative stress using hydrogen peroxide (H_2O_2) treatment. Samples were incubated for 1 week at +37°C with various concentrations of H_2O_2 (0.01 µg/mL, 0.1 µg/mL, 1 µg/mL, 10 µg/mL, 100 µg/mL and 1 mg/mL). Stock solutions of H_2O_2 were prepared at 33.3x concentrate, and 30 µl was mixed with 970 µl of the vaccine sample to give the final concentrations required. Vaccine samples were also prepared with water instead of H_2O_2 as a control and incubated at +4°C or +37°C. As with the heat-stress study, samples were tested in the ELISA with and without desorption. The whole vaccine control sample prepared with water and incubated at +4°C was included on every plate to act as a reference. The reference was tested in duplicate columns, and all other samples were tested in a single column.

2.8 Sandwich ELISA

ELISA method

For the optimized ELISA, plates were coated overnight at +4°C with 100 µL/well of anti-tetanus mAb TT010 diluted to 2.5 µg/mL (rat IgG concentration) in carbonate buffer (0.05 M, pH 9.6). The plates were washed (3x) by immersion in PBS (pH 7.4) containing 0.05% (vol/vol) Tween 20 (PBST), then blocked with 150 µL/well of PBST containing 5% (wt/vol) dried skimmed milk powder for 1 hour at +37°C. Following a second wash in PBST, serial three-fold dilutions of the samples in PBST containing 0.5% (wt/vol) dried skimmed milk powder (sample buffer, SB) were prepared in the plate (final volume 100 µL), and the plates were incubated at +37°C for 2 hours. Plates were washed as previously described, and 100 µL/well of mAb 8E1-1H1.2.1 diluted to 1.25 µg (mouse IgG concentration) in SB was added and incubated for 2 hours at +37°C. After further washing, bound 8E1-1H1.2.1 was detected using 100 µL/well of goat anti-mouse horse radish peroxidase (HRP) conjugate (Abcam Ab97040) diluted 1/4000 in SB. After a further incubation of 1 hour at +37°C and a final wash, 100 µL/well of substrate solution containing 0.5 mg/mL 2,2-azino-bis (3-ethylbenzothiazoline-6-sulfonic acid) and 0.008% H_2O_2 in 0.05 M citric acid buffer (pH 4) was added.



The reaction was allowed to develop at room temperature for up to 30 min, and the optical density (OD) was measured at 405 nm using a Molecular Devices microplate reader. When applicable, relative antigen estimates were calculated using sigmoidal curve analysis with a log transformation of the assay OD response using CombiStats software (version 6.1, EDQM).

ELISA development and optimization

Checkerboard titrations of the coating mAb (4-0.03125 µg/mL rat IgG) and the detecting mAb (0.5-0.0156 µg/mL mouse IgG) were performed to identify optimal concentrations to use in the ELISA. The mAbs were titrated with a fixed concentration of TTxd (0.276 Lf/mL) to give the signal and no TTxd (SB only) to give the background. Several anti-mouse HRP conjugate antibodies, selected for their minimal crossreactivity to the opposite species, were tested. Different blocking and sample buffer conditions were also investigated to optimize the signal-to-noise ratio, including the use of bovine serum albumin (BSA) or different concentrations of skimmed milk powder as an irrelevant protein.

ELISA qualification

A representative human vaccine product (HuB, DTaP, Al(OH)₃ adjuvant) and a veterinary vaccine product from manufacturer B (B/ Multivalent 1) were used to determine the intermediate precision of the assay and to define suitable assay validity criteria. Two batches of each product were selected, one assigned as the reference and the other assigned as the test vaccine. Samples were titrated in duplicate on a plate with two replicate plates included per assay. A column containing SB only was also included on each plate to measure the background signal. The assay was repeated on four separate days with two operators performing two assays each. The DTaP sample from HuB was tested in an additional assay due to one assay showing higher than usual variability. Individual relative estimates for the test batch against the reference batch on the same plate were calculated. From this, geomean relative estimates were calculated, and the assay precision was expressed using geometric coefficients of variation ($GCV = \{10^s - 1\} \times 100\%$ where “s” is the standard deviation of the log₁₀ transformed estimates). The geomean background OD, correlation coefficient |R|, 90% confidence intervals for the slope ratio, and 95% confidence limits for the relative estimates were recorded for each plate.

2.9 Consistency testing

The suitability of the ELISA for monitoring the batch-to-batch consistency of different lots of the same product was assessed by testing multiple lots of one vaccine product from HuA (DTaP-IPV-HepB-Hib, Al(OH)₃ adjuvant) and multiple lots of two vaccine products from HuB (dTaP and dTaP-IPV, both containing an Al(OH)₃ and AlPO₄ adjuvant). All vaccine lots were titrated in duplicate in a single assay. For each product, one lot was selected at random to be used as a product-specific reference and assigned an arbitrary “potency” value of 1. Results for all other batches were expressed relative to this product-specific reference. For the HuB vaccines, results for the dTaP-IPV product were also expressed relative to the dTaP reference batch (also assigned an arbitrary

“potency” value of 1). From this, a geomean relative estimate was calculated for each product, and the variability between individual estimates was determined (expressed as GCV, as described above).

2.10 Transfer studies

The mAb sandwich ELISA method was transferred to four industry partners (HuA, HuB, veterinary company B and veterinary company D) across five different sites. Protocols (including plate layouts) and critical reagents were provided by NIBSC, and the same vaccine samples were tested at both the receiving site (industry partner) and donating site (NIBSC). At least one representative vaccine product was included in each transfer study, and two products were included in three of the studies. For each product, two batches were tested (one assigned as the reference and the other as the test sample) so that relative antigen estimates could be calculated. Each lab performed three independent assays (two plates per assay), and all raw data was returned to NIBSC for analysis and comparison.

2.11 Curve analysis

Curve analysis was performed for seven different products (five from HuA and two from HuB) against different tetanus antigen-containing materials that could potentially serve as a reference preparation in the ELISA: The materials evaluated included WHO standards 16/302 (non-adsorbed toxoid) and 08/218 (adsorbed toxoid), manufacturer-matched drug substance (TTxd or adsorbed TTxd), and a range of manufacturer-matched drug products. All samples were titrated in duplicate, and assays were repeated on three different days. Individual curves were analyzed using a sigmoidal curve model with log transformation of the response. The slope and asymptotes for each sample were extracted from CombiStats, and the values for the final vaccine products were compared to the different reference materials on the same plate (ratio calculated for the slopes and difference calculated for the asymptotes).

3 Results

3.1 ELISA optimization

Suitable concentrations of the coating and detecting mAbs were identified from checkerboard titration assays with a fixed concentration of toxoid. 2 µg/mL TT010 and 0.125 µg/mL 8E1-1H1.2.1 was selected, which consistently gave a background OD < 0.2, a signal OD > 1, and was in excess (showed a saturating response) to help ensure a robust assay (data not shown). A conjugate antibody (Abcam, Ab97040) that had been depleted using rat immunosorbents gave a significantly lower background OD and a higher signal-to-noise ratio than other similar reagents tested and was therefore chosen as the secondary detecting antibody. PBST containing 1% (wt/vol) BSA for the blocking buffer and sample buffer significantly increased the background OD and reduced the signal-to-noise ratio compared to PBST containing skimmed milk powder (data not shown). PBST with 5% (wt/vol) milk powder for the blocking buffer and PBST with 0.5% milk powder for

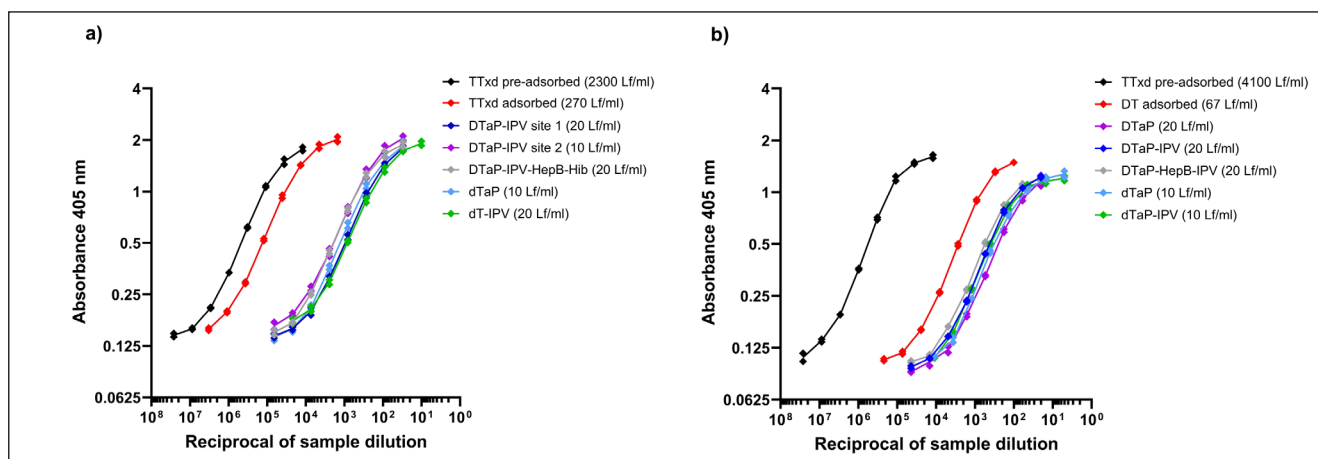


Fig. 1: Concentration response curves for drug substances and drug products from (a) HuA and (b) HuB

All samples were titrated in duplicate. Graphs show the individual data points for each replicate, with the connecting line through the mean, for a single representative assay.

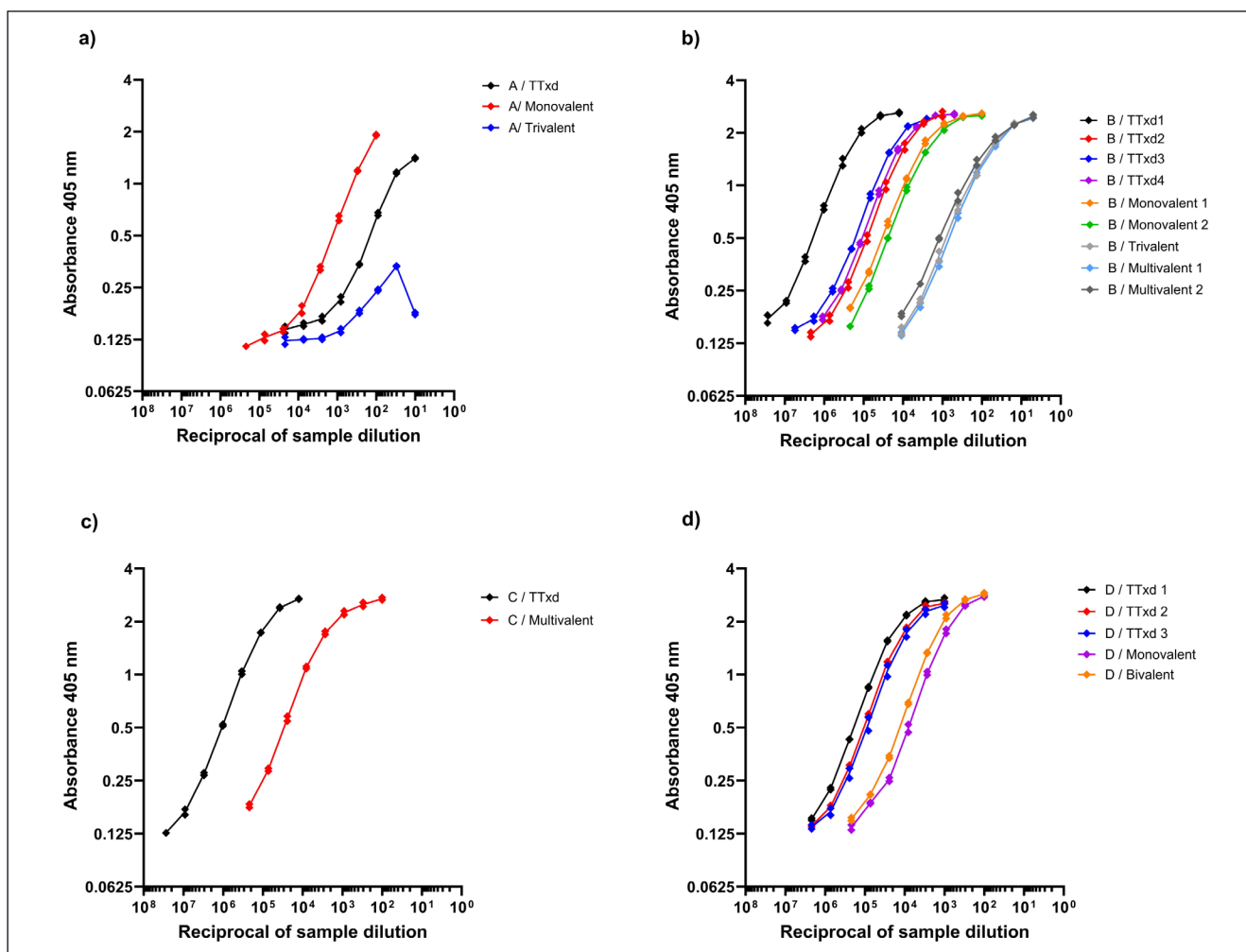


Fig. 2: Concentration response curves for drug substances and drug products from (a) veterinary company A, (b) veterinary company B, (c) veterinary company C, and (d) veterinary company D

All samples were titrated in duplicate. Graphs show the individual data points for each replicate, with the connecting line through the mean, for a single representative assay.

**Tab. 2: Proportion of TTxd detected in the presence of adjuvant compared to after a desorption step**

The degree of adsorption measured for each product is also shown. Data is the result of a single assay.

Manufacturer	Product	Adjuvant	Proportion of antigen detected (%) ^a	Degree of adsorption (%) ^b
HuA	dTaP	AlPO ₄	98	84
	DTaP-IPV-HepB-Hib	Al(OH) ₃	88	12
HuB	dTaP	Al(OH) ₃ + AlPO ₄	103	100
	DTaP	Al(OH) ₃	59	100
Veterinary company A	A / Monovalent	KAl(SO ₄) ₂	37	74
Veterinary company B	B / Trivalent	Al(OH) ₃	35	100
	B / Multivalent 1	Al(OH) ₃	47	90
	B / Multivalent 2	KAl(SO ₄) ₂	39	100
Veterinary company D	D / Monovalent	AlPO ₄	92	21
	D / Bivalent	AlPO ₄ , Quil A ISCOMS	107	14

^a (whole vaccine/(desorbed fraction + non-adsorbed fraction)) x 100; ^b (desorbed fraction/(desorbed fraction + non-adsorbed fraction)) x 100

the sample buffer were therefore chosen to give the lowest background without impacting the signal.

3.2 Assay performance characteristics

3.2.1 Use of the ELISA with different drug substances and drug products from different manufacturers

In total, twenty different final vaccine products containing a range of adjuvant types, and thirteen drug substances from six manufacturers were tested in the mAb ELISA. Vaccine samples were tested in the presence of their adjuvant (which we refer to as testing “whole vaccine”). Suitable sigmoidal concentration response curves were obtained in the ELISA for all samples tested except for the trivalent product from veterinary manufacturer A, which was low and did not give a good sigmoidal-shaped curve (Fig. 1, 2). Other lots of the same product showed a similar or lower response (data not shown). Overall, the magnitude of the response correlated well with the type of sample titrated, with the highest response generally seen for the drug substance samples (which are concentrated) compared to the final vaccine products (Fig. 1, 2).

3.2.2 Proportion of antigen detected in the whole vaccine

For vaccine products containing an aluminium adjuvant, the amount of antigen detected in the whole vaccine was compared to the amount detected after a desorption step, to estimate the proportion of antigen detected in the presence of the adjuvant. Antigen detection for the veterinary trivalent vaccine from manufacturer A was significantly improved when desorbed compared to the whole vaccine, which resulted in a very low response curve. Desorption also increased the amount of tetanus antigen available for antibody detection in the other vaccine samples tested. The estimated proportion of antigen detected in the presence of adjuvant (i.e., in the whole vaccine) was product- and adjuvant-specific (Tab. 2); for the human vaccine products tested this ranged from 59% to 100%, and for the veterinary vaccine products this ranged from 35% to 100%. In general, the proportion of antigen detected in the

whole vaccine was lower when the product contained Al(OH)₃ or KAl(SO₄)₂ adjuvant, which typically resulted in a higher degree of adsorption of antigen to adjuvant.

3.2.3 Assay precision

The intermediate precision of the ELISA method was determined by testing a human and a veterinary vaccine product (one reference batch and one test batch for each) across duplicate plates on several days. One relative antigen estimate for the human DTaP vaccine was excluded from the analysis due to wide confidence intervals around the slope ratio, and an extra assay was performed for this product. The GCV of the relative antigen estimate was 4.8% for the human vaccine product (n = 9) and 6.1% for the veterinary vaccine product (n = 8). Based on this study, validity criteria for sigmoidal curve analysis of relative antigen content were defined as follows: a weighted corre-

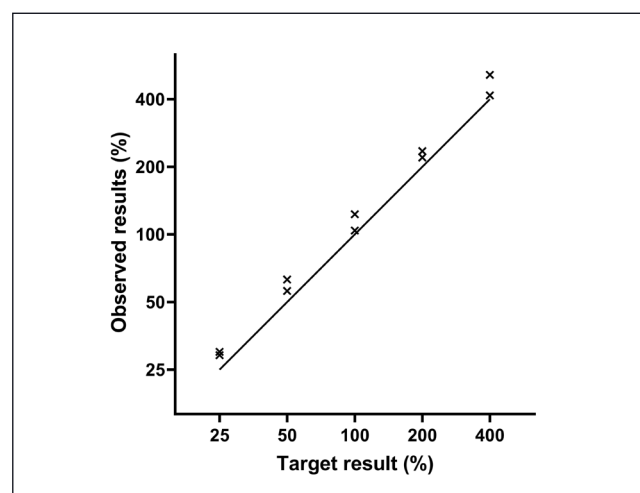


Fig. 3: Linearity analysis results for spiked dTaP samples; n = 2 assays at each target concentration level
Plot includes line of identity (slope = 1, intercept = 0).

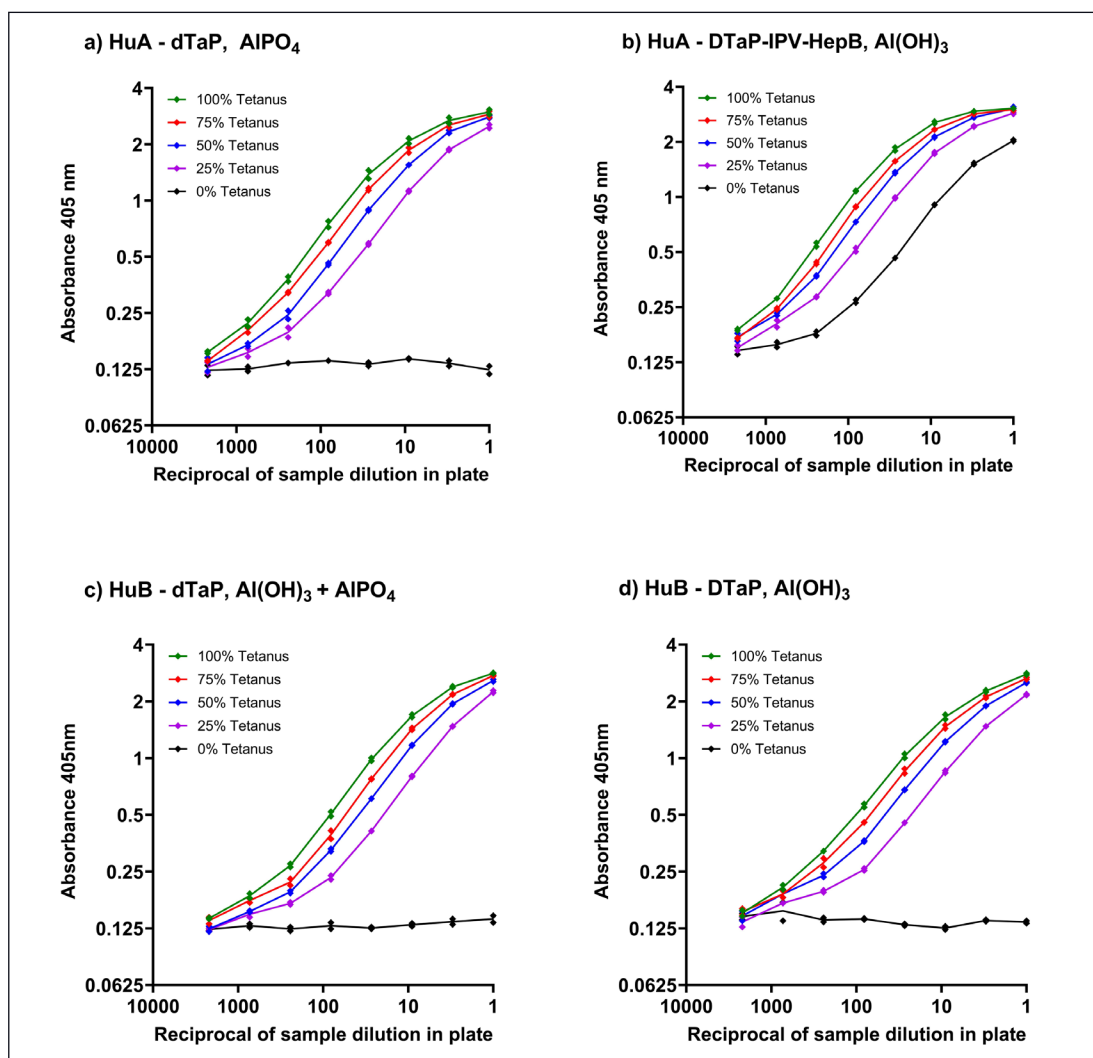


Fig. 4:
Concentration response curves for human vaccine graded dose samples from HuA (a,b) and HuB (c,d) formulated by mixing drop out samples (0% tetanus) with the normal vaccine (100% tetanus) in different ratios. Samples were tested in duplicate on a plate. Graphs show the individual data points for each replicate, with the connecting line through the mean, obtained in 1 of 2 replicate assays.

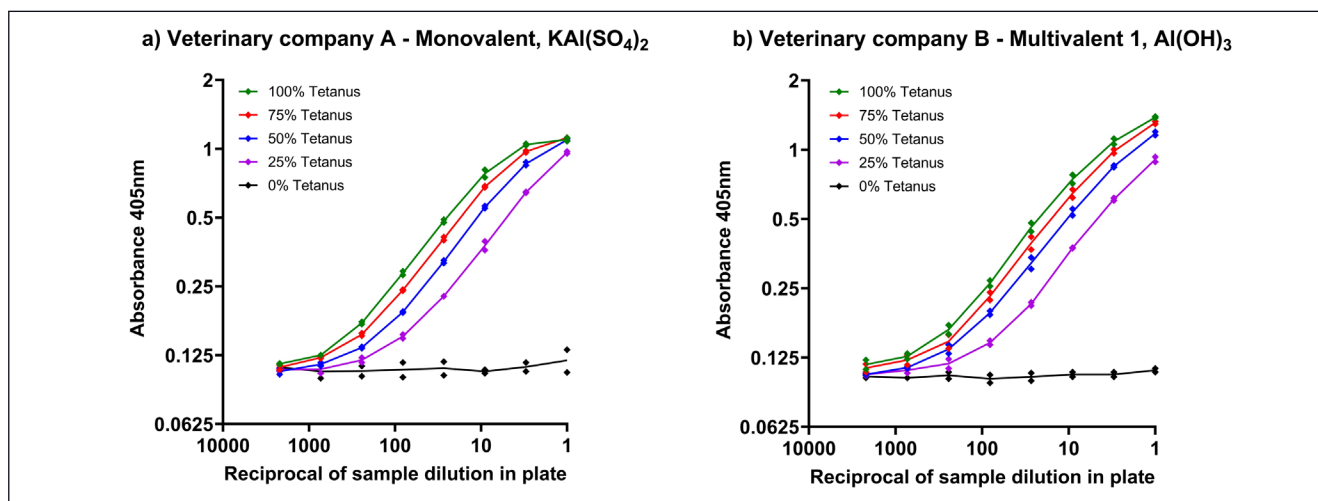


Fig. 5: Concentration response curves for veterinary vaccine graded dose samples from veterinary company A (a) and veterinary company B (b) formulated by mixing drop out samples (0% tetanus) with the normal vaccine (100% tetanus) in different ratios. Samples were tested in duplicate on a plate. Graphs show the individual data points for each replicate, with the connecting line through the mean, obtained in 1 of 2 replicate assays.

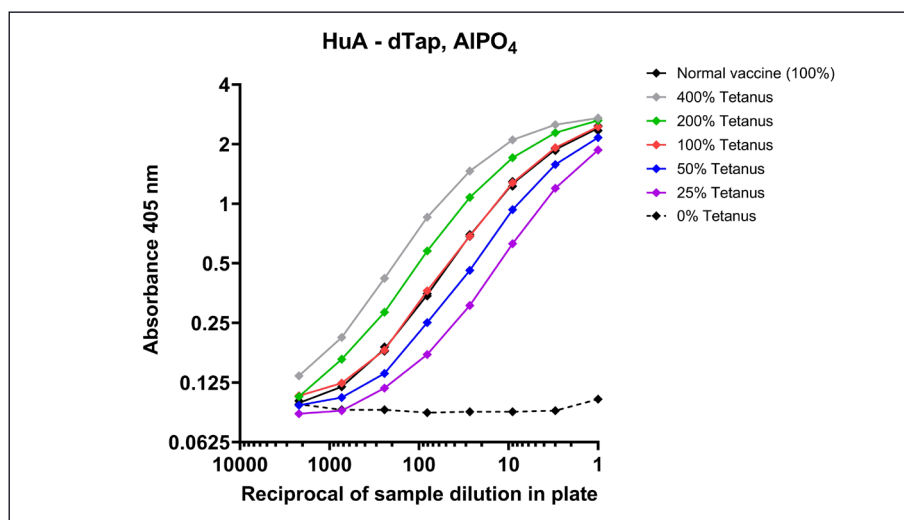


Fig. 6: Concentration response curves for spiked samples from HuA alongside a normal dTaP vaccine sample (100%)

Samples were tested in a single column except for the normal control sample, which was tested in duplicate. The graph shows the individual data points for each replicate (where applicable), with the connecting line through the mean, obtained in 1 of 2 replicate assays.

Tab. 3: Amount of antigen detected in the graded dose samples

Results are the geomean of two independent assays.

Graded dose sample	Tetanus antigen content relative to 100% sample (%)					
	HuA		HuB		Veterinary A	Veterinary B
	dTaP	DTaP-IPV-HepB-Hib	dTaP	DTaP	A/Monovalent	B/Multivalent 1
0% T	< 0.1	8	< 0.1	< 0.1	< 0.1	< 0.1
25% T	23	28	26	24	22	24
50% T	45	55	50	47	45	50
75% T	69	71	71	71	72	74

lation coefficient $|R|$ greater than 0.975, 90% confidence intervals for slope ratio within 0.85-1.18 to confirm parallelism, and 95% confidence limits for the relative antigen estimate within 80-125% of the estimate. A target intermediate precision (GCV) for the transfer studies to other laboratories was also defined as 15% or less.

3.2.4 Assay linearity and specificity

Dilutional linearity was assessed using data from the spiking study where the drop out dTaP vaccine from HuA was spiked with adsorbed TTxd to give 25%, 50%, 75%, 100%, 200% and 400% tetanus antigen (Fig. 3). Estimates for the spiked samples were calculated relative to the normal vaccine sample. There was good agreement between the observed and target values with a fitted slope not significantly different from 1.0 (0.99 with 95% CI of 0.93-1.05), an intercept not significantly different from 0 (0.09 with 95% CI -0.03 to 0.21), and a high r^2 value (> 0.99).

No signal was observed for the dTaP and DTaP TTxd drop out vaccines from the human vaccine manufacturers (Fig. 4) and the two aluminium vaccine products tested from the veterinary manufacturers (Fig. 5). For the DTaP-IPV-HepB-Hib vaccine product from HuA a signal was detected in the 0% tetanus sample (Fig. 4b); however, this was not unexpected since the Hib component in this vaccine is conjugated to a TTxd carrier protein.

3.2.5 Detection of under- and over-formulation of vaccine samples

The graded dose samples formulated by mixing the TTxd drop out vaccine samples with the normal vaccine showed a decrease in signal corresponding well to the amount of tetanus antigen present (Fig. 4, 5 and Tab. 3). Similarly, results from spiking studies with the dTaP vaccine from HuA showed a decrease in signal for the 25% and 50% samples, and an increase in signal for the 200% and 400% samples, equivalent to the TTxd concentration (Fig. 6).

3.2.6 Detection of sonication-induced alterations in non-adjuvanted TTxd

Non-adjuvanted TTxd diluted to final vaccine concentrations and subjected to sonication showed an increase in large particles (aggregates) and a general decrease in soluble antigen as the number of cycles increased (Fig. 7). The increase in aggregates correlated well with changes in the response observed in the mAb ELISA (Spearman's rank correlation coefficient $\rho = 0.964$; $p < 0.001$), which showed a decrease in antigenicity with an increasing number of sonication cycles (Fig. 8). Reducing the sonication energy or increasing the cooling interval between cycles reduced the changes observed in both HPLC and ELISA (Fig. 7, 8). Note that the extra peak observed in Figure 7b at approximately 35 minutes is buffer-derived.

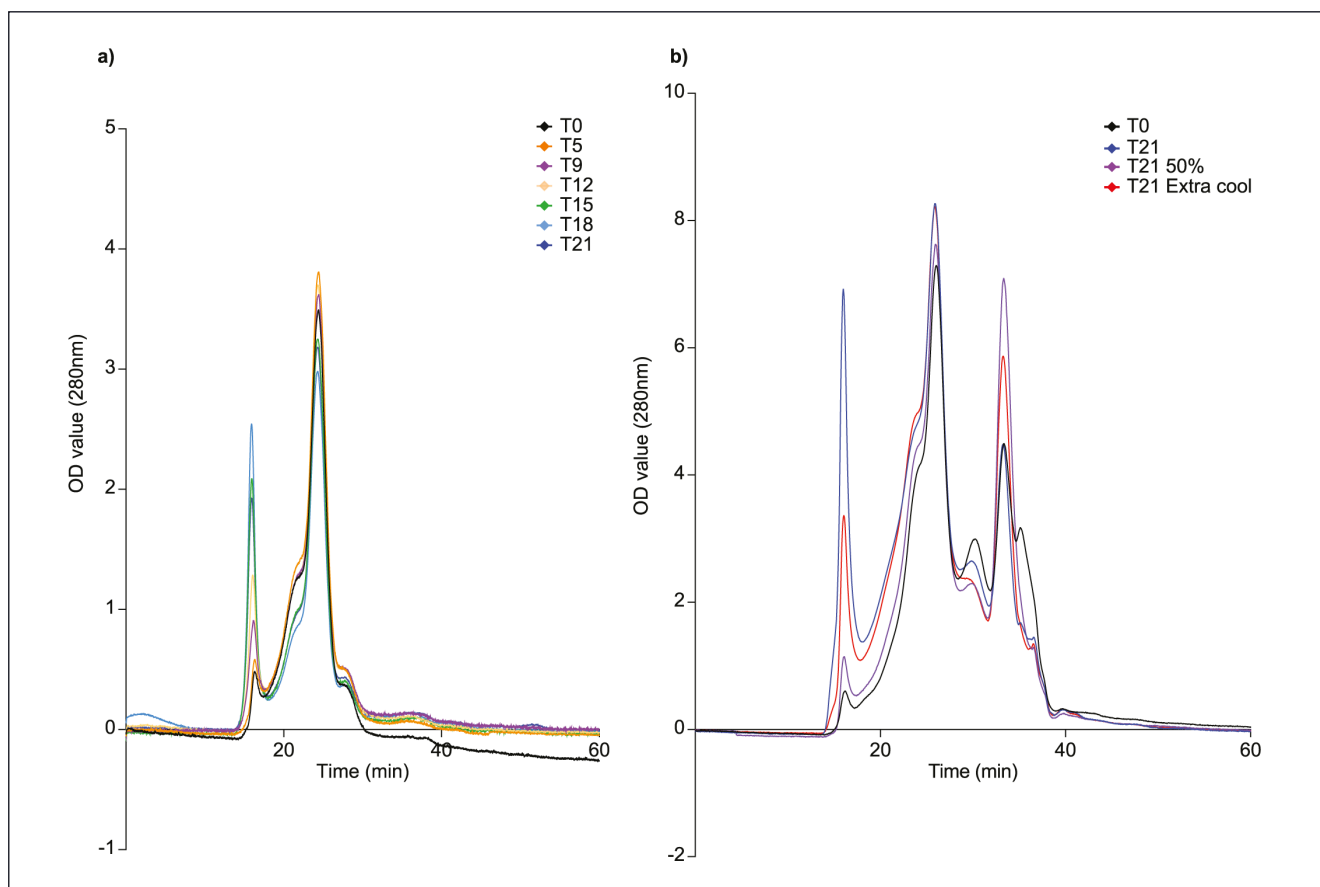


Fig. 7: Effect of sonication on the structure of TTxd

Data shows HPLC chromatograms obtained for TTxd (a) subjected to an increasing number (T0-T21) of standard cycles and (b) for the T21 sample subjected to less harsh conditions (lower power, T21 50% and increased cooled, T21 extra cool).

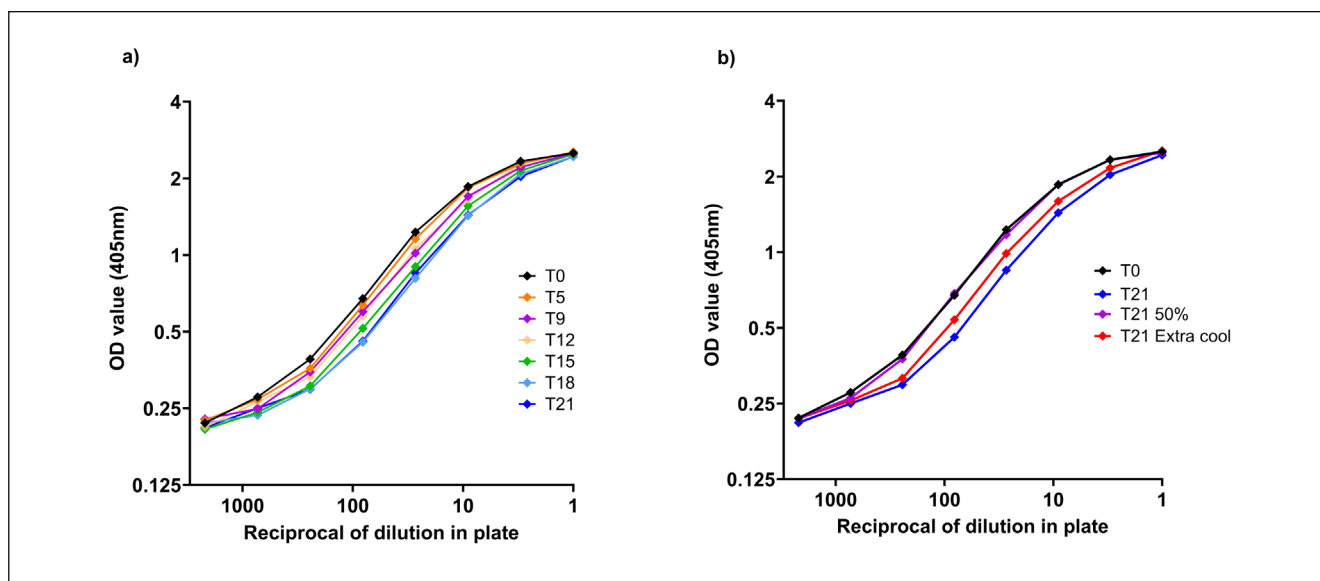


Fig. 8: Effect of sonication on TTxd antigenicity in the mAb ELISA

Data shows TTxd subjected to (a) an increasing number (T0-T21) of standard cycles and (b) the T21 sample subjected to less harsh conditions (lower power, T21 50% and increased cooled, T21 extra cool). Samples were tested in a single column on a plate in a single assay.

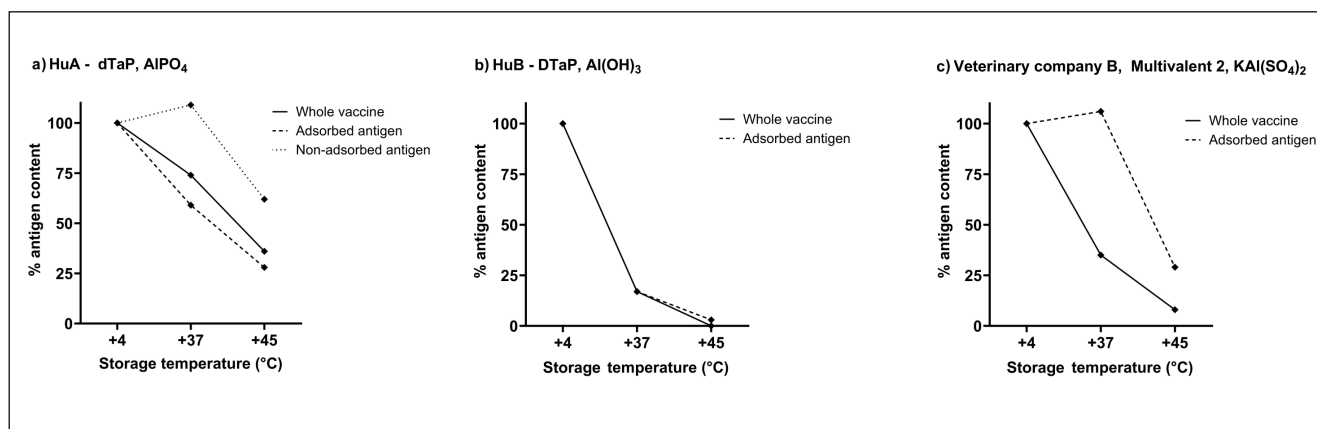


Fig. 9: Effect of storage temperature on the amount of TTxd detected in a dTaP vaccine from HuA (a), a DTaP vaccine from HuB (b), and a multivalent vaccine from veterinary company B (c)

Estimates for the samples stored at +37°C and +45°C are expressed as a percentage of the equivalent +4°C sample (whole vaccine, adsorbed antigen fraction or non-adsorbed antigen fraction). Note that no non-adsorbed antigen was detected in the product from HuB (b) or veterinary company B (c). Data is the result from a single assay. The reference sample was tested in duplicate columns, and all other samples were tested in a single column.

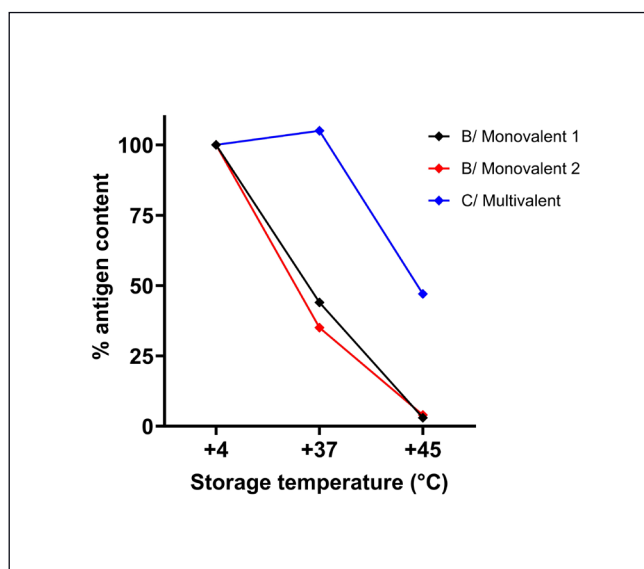


Fig. 10: Effect of storage temperature on the amount of TTxd detected in non-aluminium adjuvanted monovalent vaccines from veterinary company B and a multivalent vaccine from veterinary company C

Estimates for the samples stored at +37°C and +45°C are expressed as a percentage of the +4°C sample. Data is the result from a single assay. The reference sample was tested in duplicate columns, and all other samples were tested in a single column.

3.2.7 Detection of temperature-induced alterations in vaccine samples

A selection of aluminium adjuvanted and non-aluminium adjuvanted tetanus vaccines were subjected to heat treatment. A loss of antigen content was detected to varying degrees in all vaccine products incubated at elevated temperatures (Fig. 9, 10). For the aluminium adjuvanted samples, desorption of the incubated samples was also performed to help confirm that any changes detected in the ELISA were due to a change in antigenicity of the TTxd as opposed to being an artefact of the antigen accessibility changing (e.g., if heating caused changes in the adsorption profile). Estimates for the samples stored at elevated temperatures were expressed as a percentage of the equivalent +4°C sample (whole vaccine, adsorbed antigen fraction or non-adsorbed antigen fraction). For the aluminium adjuvanted product from HuA (dTaP, AlPO₄ adjuvant) and HuB (DTaP, Al(OH)₃) a similar loss of antigen was detected in both the whole vaccine sample and the adsorbed antigen sample (after desorption). For the aluminium adjuvanted veterinary vaccine (B/ Multivalent 2, KAl(SO₄)₂ adjuvant), there was a 65% reduction in antigen content measured for the whole vaccine sample stored at +37°C, but no loss measured for the desorbed vaccine sample. At +45°C, a loss of antigen content was detected in both samples, with the whole vaccine exhibiting a greater degree of loss.

3.2.8 Detection of chemical-induced alterations in vaccine samples

The effect of H₂O₂ (1 week at +37°C) was also assessed for a smaller selection of aluminium adjuvanted and non-aluminium adjuvanted tetanus vaccines. As with the heat stress study, aluminium adjuvanted vaccines were tested in the ELISA with and without desorption. Incubating the vaccine at +37°C for a week with no H₂O₂ caused a small loss in the antigen compared to the control sample that was stored at +4°C (data not shown). These changes are in line with those observed in the vaccine heat treat-

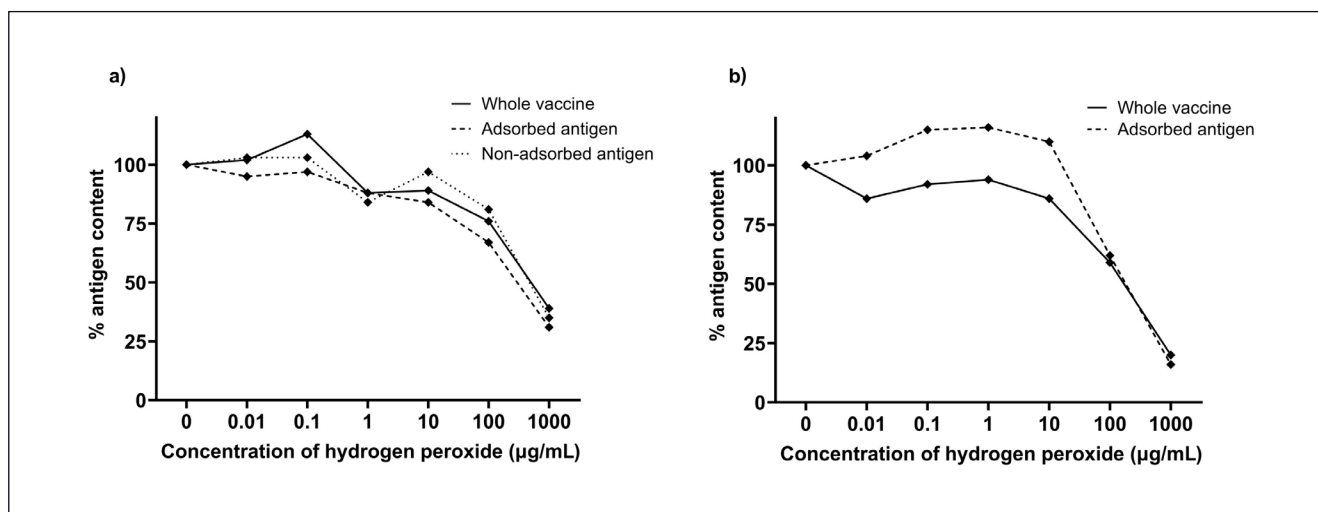


Fig. 11: Effect of hydrogen peroxide on the amount of TTxd detected in the (a) dTaP vaccine (AlPO₄ adjuvant) from HuA and (b) DTaP vaccine (Al(OH)₃ adjuvant) from HuB (b)

Estimates for the samples incubated with H₂O₂ are expressed as a percentage of the equivalent +37°C no-H₂O₂ control sample (whole vaccine, adsorbed antigen fraction or non-adsorbed antigen fraction). Note that no non-adsorbed antigen is detected in the DTaP vaccine from HuB (b). Data is the result from a single assay. The reference sample was tested in duplicate columns, and all other samples were tested in a single column.

ment studies, albeit to a lesser extent due to the shorter time-period of incubation (1 week compared to 8 weeks).

Estimates for the samples incubated with H₂O₂ were expressed as a percentage of the equivalent +37°C no-H₂O₂ control sample (whole vaccine, adsorbed antigen fraction or non-adsorbed antigen fraction). A similar trend was observed for the different fractions tested for a particular product. A concentration of 100 µg/mL H₂O₂ was required to cause a significant change in the amount of antigen detected in all three products (Fig. 11, 12).

3.2.9 Consistency testing

The individual potency estimates for the different batches tested were precise with confidence intervals within 90-107% of the estimate in all cases. A summary of the batch-to-batch variability observed is shown in Figure 13 and in Table 4. The variability between estimates (GCV) was only slightly higher than that observed during the ELISA qualification, suggesting that the ELISA will perform well with real world samples.

More antigen was detected in the dTaP-IPV product when using a heterologous reference (dTaP) compared to a homologous reference even though they are formulated with the same amount of TTxd. The lot-to-lot variability however was comparable, suggesting that use of a product-type (as opposed to product-specific) reference preparation may be suitable for the calculation of relative antigen estimates.

3.3 Transfer studies

The tetanus ELISA was successfully transferred to four industry partners (across five different laboratory sites). The receiving laboratories did not report any technical issues with method performance and only one minor deviation from the pre-defined success criteria was observed. In total, data from sixty ELISA plates was

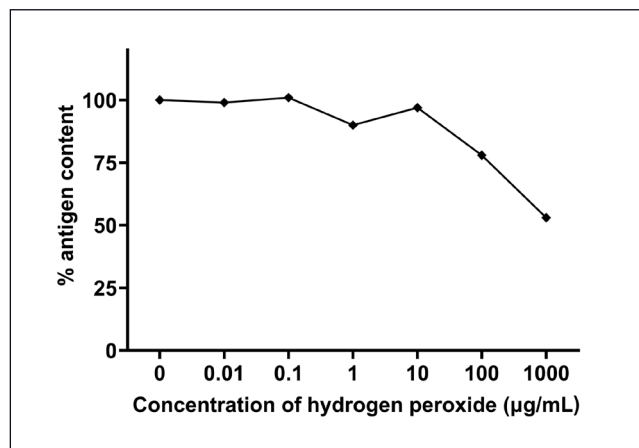


Fig. 12: Effect of hydrogen peroxide on the amount of TTxd detected in a non-aluminium adjuvanted monovalent vaccine from veterinary company B (B/ Monovalent 1)

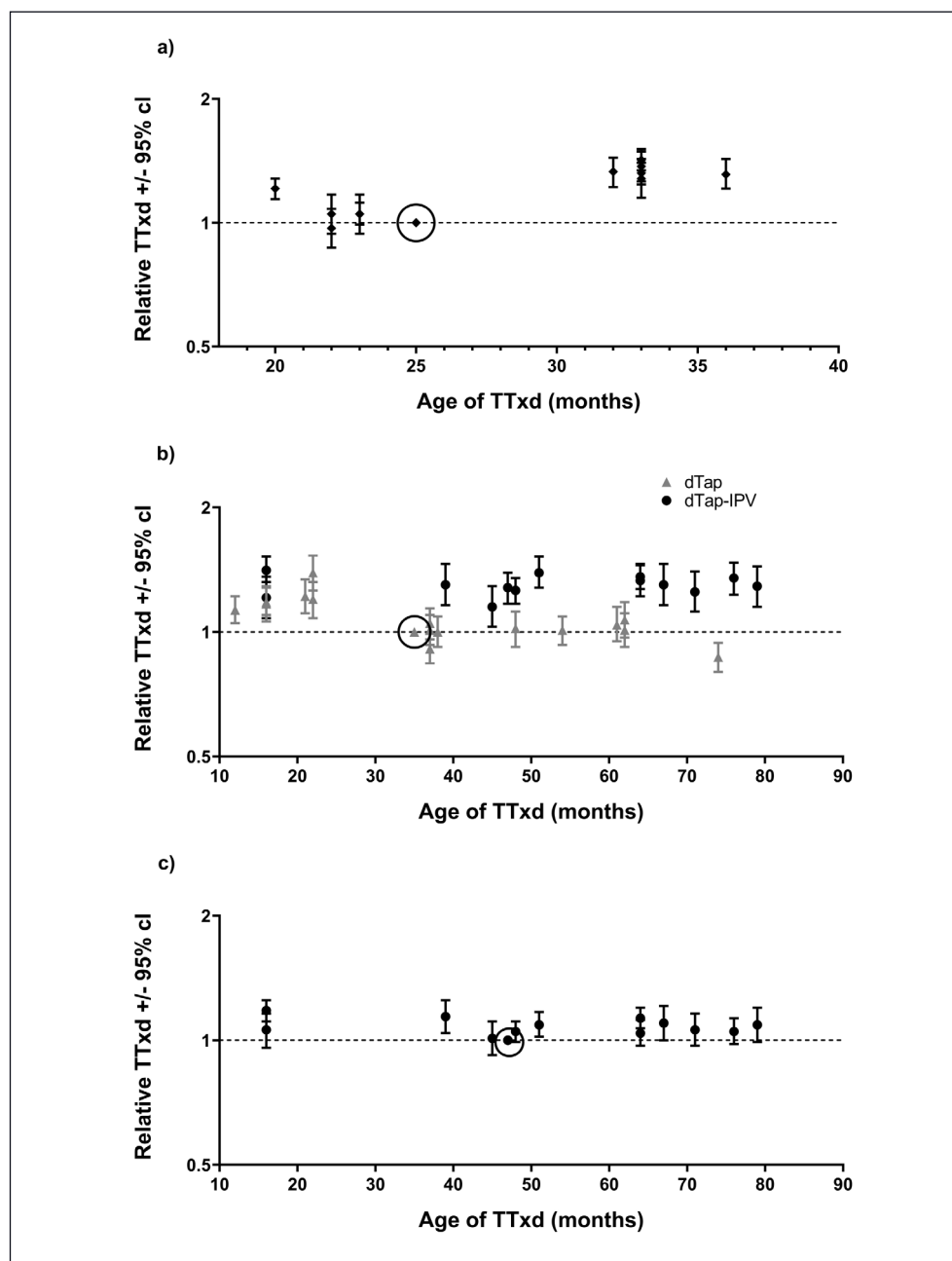
Estimates for the samples incubated with H₂O₂ are expressed as a percentage of the equivalent +37°C no-H₂O₂ control sample. Data is the result from a single assay.

analyzed by NIBSC, and only one plate failed to meet the assay validity criteria and was consequently excluded from the analysis. Target acceptance criteria for a successful transfer were defined as 1) an intermediate precision (GCV) < 15%, 2) GCV for partner lab < GCV for NIBSC +5% and 3) geomean relative potency estimate for partner lab within 10% of the estimate obtained at NIBSC. The intermediate assay precision was less than 15% for all products tested, and the GCV obtained by the partner lab was within 5% of that obtained by NIBSC. The one deviation that was



Fig. 13: Batch-to-batch variability observed for a DTaP-IPV-HepB product from HuA (graph a, black diamonds) and a dTaP (graph b, grey triangles) and dTaP-IPV (graphs b and c, black circles) product from HuB

Graphs show relative estimates calculated against a reference batch (highlighted with a circle). Results for dTaP-IPV batches against a homologous reference are shown in graph c, and results for the same batches against a heterologous dTaP reference are shown in graph b. All samples were titrated in duplicate columns in a single assay.



observed was for the DTaP-IPV-HepB-Hib product, where the geomean relative estimate for the partner lab was not within 10% of the geomean relative estimate obtained at NIBSC. Results for all studies are summarized in Table 5.

3.4 Curve analysis

Suitability of different materials to act as a reference preparation in the ELISA was assessed for seven different products by examining the similarity of the concentration-response curve shapes. Test sample and candidate reference sample curves were considered to be similar if the difference between the upper or lower asymptotes fell within -0.05 and 0.05 and if the ratio of the slopes

was between 0.90 and 1.11. In the first study two batches of each product were tested against WHO standards 16/302 and 08/218 and drug substances from the same manufacturer (TTxd and adsorbed TTxd) on the same plate. The geomean slope ratio and average asymptote differences for 3 replicate assays are summarized in Table 6. Small differences were observed in the upper asymptotes and slopes for some of the products compared to WHO standards, but generally not against the drug substances from the same manufacturer. In the second study different products from the same manufacturer were tested against each other. Products from HuA were split into 2 different “families” from different manufacturing sites, and only products in the same family were

Tab. 4: Batch-to-batch consistency monitoring of three different products

Results show relative estimates calculated against a product-specific homologous reference or against a “product-type” heterologous reference (dTaP-IPV against dTaP only). Data is the result of a single assay.

Manufacturer	Product	Reference type	Number of batches	Geomean (95% CI)	GCV (%)
HuA	DTaP-IPV-HepB-Hib	DTaP-IPV-HepB-Hib	11	1.10 (1.03-1.18)	10.8
HuB	dTaP	dTaP	16	1.08 (1.02-1.15)	10.5
	dTaP-IPV	dTaP	13	1.55 (1.47-1.65)	10.0
	dTaP-IPV	dTaP-IPV	12	1.14 (1.07-1.20)	9.1

Tab. 5: Results of transfer studies with industry partners

Geomean relative estimates and GCVs for each lab were calculated from results obtained in 3 independent assays (n = 2 per assay).

Manufacturer	Product	Geomean relative estimate difference (%)	Assay precision, GCV (%)		
			Partner	NIBSC	Difference
HuA site 1	DTaP-IPV-HepB-Hib	14.2	3.4	4.4	-1.0
HuA site 2	dTap	1.6	4.5	3.8	0.7
HuB	dTaP	5.5	2.7	3.9	-1.2
	DTaP	4.4	7.2	3.3	-3.9
B	B / Multivalent 3 ^a	4.8	12.3	13.5	1.2
	B / Multivalent 1	9.6	5.3	7.0	1.7
D	D / Multivalent 1 ^a	4.6	1.8	1.9	0.1
	D / Bivalent ^a	3.6	4.6	6.6	2.0

^a Only provided for the transfer studies, not tested previously

tested against each other. No differences in the curve shapes were observed for any of the products assessed (data not shown), suggesting that products from the same manufacturer would be suitable to act as references for each other.

4 Discussion

Antigen identity, quantity and integrity are key factors to be evaluated as part of consistency testing of tetanus vaccines. Several ELISA methods have been described previously for the qualitative and quantitative characterization of TTxn and TTxd, both during production and in final preparations of tetanus vaccines (Prieur et al., 2002; Burkin et al., 2004; Šeatović et al., 2004), and we have demonstrated suitability of an in-house sandwich ELISA for measuring TTxd in both human and veterinary tetanus vaccines (Coombes et al., 2012; Riches-Duit et al., 2019). For implementation of a quality control test the design of an assay needs to reflect both antigen content and functionality. As such, an ELISA should use well-characterized antibodies that are of high affinity and that target epitopes relevant to the protection offered by the vaccine and which are stability indicating (Council of Europe, 2018). The previous in-house ELISA that we developed used a pAb for detection of the captured antigen, which is harder to characterize and requires the use of animals every time a new batch is produced.

Therefore, within the VAC2VAC project we performed an extensive characterization study on a panel of four mAbs (including affinity, functional activity, and recognition of heat-altered antigen) and selected a suitable pair to develop into a sandwich ELISA (Riches-Duit et al., 2021).

Here we have optimized the mAb ELISA method and evaluated suitability of the method as a consistency test. We have confirmed the specificity of the assay for TTxd by testing drop out samples for a range of representative vaccine products that contain all components of the vaccine except the tetanus toxoid antigen. No response was observed for any of the drop out samples except the one containing a Hib component conjugated to TTxd as a carrier protein. We have previously demonstrated that TTxd conjugated to polysaccharide could be detected in a similar sandwich ELISA assay, but that the amount of conjugated TTxd detected was considerably lower than conventional TTxd (Coombes et al., 2012). Detection of the carrier protein in conjugate vaccines containing tetanus toxoid will not adversely affect the use of the assay described here as a consistency tool.

The final vaccine products tested varied in their composition in terms of TTxd concentration, the type and number of other non-tetanus antigens present, and adjuvant type (aluminium based or non-aluminium based). For the majority of products, the ELISA can be applied to the whole vaccine sample without the need for any pre-treatment (i.e., in the presence of the adjuvant). All whole vac-



Tab.6: Comparison of curve shapes for different products against WHO standards and drug substances from the same manufacturer

Slope ratios < 0.90 or > 1.11 and differences in asymptote of < -0.05 or > 0.05 are highlighted in grey. Data shows the geomean slope ratio and average asymptote differences from 3 independent assays.

Manufacturer	Product	Batch	Slope ratio				Lower asymptote difference				Upper asymptote difference			
			16/302	08/218	TTxd	Ad-sorbed TTxd	16/302	08/218	TTxd	Ad-sorbed TTxd	16/302	08/218	TTxd	Ad-sorbed TTxd
HuA	DTaP-IPV-HepB-Hib	1	0.962	0.970	0.973	1.054	-0.015	-0.007	-0.003	0.006	0.026	0.021	0.011	-0.050
		2	0.959	0.967	0.970	1.051	-0.017	-0.010	-0.006	0.004	0.036	0.031	0.021	-0.039
	DTaP-IPV	1	0.930	0.962	0.961	1.004	-0.013	0.003	0.007	0.003	0.033	0.020	0.013	-0.037
		2	0.940	0.973	0.972	1.016	-0.013	0.003	0.007	0.003	0.040	0.027	0.020	-0.030
	dT-IPV	1	0.968	0.968	0.986	1.053	-0.005	0.002	0.017	0.010	0.032	0.028	0.017	-0.042
		2	0.989	0.989	1.007	1.075	0.006	0.012	0.027	0.021	0.028	0.023	0.012	-0.047
	Tdap	1	0.937	0.946	0.973	1.007	-0.004	0.005	0.012	0.014	0.064	0.054	0.015	0.003
		2	0.895	0.903	0.929	0.962	-0.017	-0.009	-0.002	0.000	0.085	0.075	0.036	0.023
	DTaP-IPV	1	0.960	0.940	0.985	0.972	-0.009	-0.016	-0.002	-0.013	0.073	0.065	0.012	0.022
		2	0.987	0.966	1.013	0.999	0.002	-0.005	0.008	-0.002	0.065	0.058	0.005	0.014
HuB	dTaP	1	0.865	0.888	0.880	0.995	-0.029	-0.021	-0.032	0.022	0.053	0.042	0.035	0.030
		2	0.925	0.950	0.941	1.064	0.003	0.011	0.000	0.054	0.039	0.028	0.021	0.016
	DTaP	1	0.911	0.938	0.932	1.000	-0.027	-0.019	-0.015	-0.002	0.016	0.006	0.006	-0.007
		2	0.903	0.931	0.924	0.991	-0.018	-0.010	-0.006	0.007	0.013	0.003	0.003	-0.010

cine samples gave good concentration response curves except the trivalent product from veterinary manufacturer A, which needed to be desorbed to obtain a good concentration-response curve. Validation of a desorption step will increase complexity of the method but at the same time provide additional information with regards to the degree of adsorption, which can and should be monitored on a batch-to-batch basis. Where desorption is not strictly required, the adsorption profile of a product could still be assessed by simply centrifuging the vaccine sample and including the supernatant in the ELISA to monitor the relative amount of non-adsorbed TTxd present in each lot.

Adsorption onto an aluminium adjuvant can mask the epitopes on the TTxd, either preventing (as in the case of the trivalent product from veterinary manufacturer A) or reducing binding of the mAbs to their target epitopes. To investigate this further we performed desorption studies with products adsorbed onto an aluminium adjuvant and compared the amount of TTxd detected after a desorption step to that detected in the whole vaccine sample. For the veterinary products containing an $\text{Al}(\text{OH})_3$ or $\text{KAl}(\text{SO}_4)_2$ adjuvant, the proportion of antigen detected in the whole vaccine sample was approximately 40%, whereas most or all of the antigen was detected in the whole vaccine samples containing an AlPO_4 adjuvant. The proportion of antigen detected correlates well with the adsorption capacity of the adjuvant and the degree of adsorption measured; AlPO_4 has a lower adsorption capacity, and a higher proportion of the antigen was detected in these vaccine samples where less of the TTxd was adsorbed. For the human vaccine products, the proportion of tetanus antigen detected ranged from 59-100%. As with the veterinary samples, the proportion of antigen detected

was higher when the degree of adsorption was low. However, this was affected by not only the adjuvant type but also the other non-tetanus antigen components present in the vaccine sample. It should be noted that the desorption step may not be 100% efficient and that the effect of the desorption process on the antigen is not fully understood. However, we have demonstrated that the desorption conditions themselves (i.e., as applied to a non-adjuvanted tetanus toxoid sample) do not affect the antigenicity measured by ELISA (data not shown). In terms of monitoring consistency of production, it may not be necessary for 100% of the antigen to be detected as long as the proportion profile that is detected over the vaccine's shelf life is consistent and representative of the quality of the vaccine as a whole. This will need to be evaluated on a product-specific basis.

Assays using graded dose samples and spiked samples were performed to investigate the dilutional linearity of the assay and to determine if formulation errors would be detected accurately. The graded dose samples gave the expected pattern of response for the human and veterinary vaccine products tested and showed that the ELISA can detect 25% changes in antigen content, something that would be difficult to see using an animal assay (Stalpers et al., 2021). Spiking studies also confirmed that the assay has good dilutional linearity (covering at least the range of 25-400% of a booster dose human vaccine).

The potency assay has traditionally been used as a measure of stability to detect changes over time in the potency of a product. We have demonstrated that the mAb ELISA is able to discriminate between normal and stressed samples of non-adjuvanted TTxd (after exposure to sonication), and between normal and stressed final

lot vaccine (after exposure to heat or hydrogen peroxide). Sonication caused changes in the structure of TTxd that were detected by HPLC, namely an increase in the large particles detected and a decrease in the soluble antigen. The ELISA correlated well with HPLC results, with a decrease in antigen detected that was dependent on the number of sonication cycles, the sonication energy, and the cooling interval. A reduction in signal for all samples was detected after exposure to elevated temperatures and high concentrations of H_2O_2 . The adjuvanted samples were tested in the ELISA with and without desorption to confirm that any loss in signal we observed was not just due to a change in the adsorption profile affecting the accessibility of the mAb epitopes on the toxoid. For the aluminium adjuvanted products from HuA (dTAP, AlPO_4 adjuvant) and HuB (DTAP, $\text{Al}(\text{OH})_3$), a similar pattern of loss was observed in the whole vaccine sample and the corresponding adsorbed antigen sample (after desorption), suggesting a real decrease in antigenicity. For the heat-treated aluminium adjuvanted veterinary vaccine (B/ Multivalent 2, $\text{KAl}(\text{SO}_4)_2$ adjuvant), there was a 65% reduction in antigen content measured for the whole vaccine sample stored at $+37^\circ\text{C}$, but no loss measured for the adsorbed antigen sample. This suggests that, in the whole vaccine sample, incubation at the elevated temperature is causing a change in antigen accessibility rather than decreasing antigenicity. At the higher temperature of $+45^\circ\text{C}$, where a loss of antigen content is observed in both samples but to a larger extent in the whole vaccine, the observed reduction in binding is likely to be caused by both a change in antigen accessibility and degradation.

In-house qualification assays confirmed that the mAb ELISA has very good reproducibility with an intermediate precision (GCV) of 5–6%. Furthermore, we tested multiple lots of three different human vaccine products covering a range of different “ages” (from the date of manufacture of the final bulk to the date of testing) from a few months to several years. The between-batch variability observed was only slightly higher (GCV of approximately 9%) than the within-batch variability, demonstrating that the assay should work well with real world samples and be suitably powerful for detecting quality differences between different vaccine batches.

In most regions, potency assays for tetanus vaccine are standardized by expression of potency estimates relative to a reference vaccine that is traceable to the WHO IS for Tetanus Vaccine, Adsorbed (Tierney et al., 2011). This has enabled minimum criteria for vaccine potency in IU to be defined in regulatory guidelines and monographs (WHO, 2005; Council of Europe, 2008). A move away from these *in vivo* potency assays to the use of *in vitro* assays as part of a consistency approach will almost certainly require development of product-specific acceptance criteria that will enable manufacturers to demonstrate production of vaccine batches that are consistent with batches shown to be safe and effective in clinical studies or routine use. Here we have assessed the suitability of different materials (including WHO ISs and manufacturer-specific drug substances and drug products) to act as a reference preparation for calculation of relative antigen estimates. For relative analysis, concentration-response curves for the test sample and reference should be equivalent. For example, for sigmoidal curve analysis, the curves should share common functional parameters (i.e., Hill slope, upper and lower asymptotes). In this study, manufacturer drug substances had more comparable curve shapes to fi-

nal drug products compared to the ISs, and specific drug products were suitable to act as a reference preparation for other products produced by the same manufacturer. When assessing the use of the assay as a consistency test, we also evaluated the ability of a heterologous product from the same manufacturer to be used as a reference. The lot-to-lot variability was very comparable to that observed when using a homologous reference, also suggesting that a product-type reference from the same manufacturer (as opposed to product-specific) may be suitable for the calculation of relative antigen estimates. More antigen was detected in the dTaP-IPV product when the heterologous dTaP reference was used, and this may be due to the additional IPV component affecting the adsorption profile and increasing the accessibility of the tetanus antigen to the mAbs in the dTaP-IPV product compared to in the dTaP product (as seen in the desorption study where the degree of adsorption with a particular adjuvant varied depending on the other non-tetanus antigens present). Such observations again indicate that the antigen ELISA, while quantitative, will not provide true estimates of antigen content but can provide precise estimates of relative antigen content for the purposes of monitoring batch-to-batch consistency.

Collaboration between all stakeholders is paramount to achieving progress toward regulatory acceptance of alternative, non-animal methods. Of equal importance is the need for international harmonization of regulatory requirements to help accelerate acceptance and remove the hurdle of different regions having different expectations (Lilley et al., 2023; Viviani et al., 2023). Assay development is only the first step of the transition from *in vivo* to *in vitro* batch testing, which has to be complemented by a thorough validation process. Here, we have shown proof of concept for the mAb ELISA approach for human and veterinary tetanus vaccines, including demonstration that the assay can be successfully transferred to other laboratories. The monoclonal antibodies used in this assay are publicly available for the purposes of further developing and validating this assay for routine quality control in a Good Manufacturing Practice environment.

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Conflict of interest

Authors have no conflicts of interest.

Data availability

Data described in the manuscript can be obtained from the corresponding author upon request.

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